

PAC-Bayesian Learning

Detailed solutions to the exercises of the lecture notes

Master MALIA – Ensemble Methods

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This document gathers detailed solutions to the exercises that close each section of the lecture notes *PAC-Bayesian Learning: A Theoretical Framework for Ensemble Methods*. Exercises, figures, equations and results are numbered as in the notes. The numerical values were computed with `pbttools.py`; the short scripts used for the simulation exercises are those of the notebooks. We advise to try each exercise before reading its solution: most of them take a few lines once the right tool of the notes has been identified, and the remarks that follow the solutions point to what the exercise is meant to teach.

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Section 1 – Why PAC-Bayes for ensemble methods?

Exercise 1.1

Write the predictors of bagging, of AdaBoost and of a kernel machine with random Fourier features in the form (1.1). In each case, what is the set of voters $\mathcal{H}$, and what is the distribution Q ?

Solution. In all three cases we look for a set of voters $\mathcal{H}$ and a probability distribution Q on $\mathcal{H}$ such that the predictor equals $\text{sign}(\mathbb{E}_{h \sim Q} h(x))$.

Bagging and random forests. The voters are the n trees $h_1, \dots, h_n$ learned on the bootstrap samples, so $\mathcal{H} = \{h_1, \dots, h_n\}$, and Q is the uniform distribution $Q(h_t) = \frac{1}{n}$. Indeed $\text{sign}(\frac{1}{n} \sum_t h_t(x)) = \text{sign}(\mathbb{E}_{h \sim Q} h(x))$.

AdaBoost. The voters are the weak learners $h_1, \dots, h_T$, and $Q(h_t) = \alpha_t / \sum_s \alpha_s$. The weights are non-negative because each weak learner has a weighted error $\varepsilon_t < \frac{1}{2}$, so that $\alpha_t = \frac{1}{2} \ln \frac{1-\varepsilon_t}{\varepsilon_t} > 0$, and dividing the score $\sum_t \alpha_t h_t(x)$ by the positive constant $\sum_s \alpha_s$ does not change its sign. If some α_t were negative, we would replace h_t by $-h_t$ and α_t by $|\alpha_t|$: a vote with signed weights is always a vote with non-negative weights over the set $\mathcal{H} \cup (-\mathcal{H})$.

Random Fourier features. The kernel machine is $\text{sign}(\sum_{k=1}^K \alpha_k \cos(\omega_k^\top x + b_k))$. The voters are the functions $h_{\omega, b, s}(x) = s \cos(\omega^\top x + b)$ with $s \in \{-1, +1\}$ (which take values in $[-1, 1]$ rather than $\{-1, +1\}$; for binary voters take $s \text{ sign} \cos(\omega^\top x + b)$ as in Fig. 32), and Q puts the weight $|\alpha_k| / \sum_j |\alpha_j|$ on the voter $(\omega_k, b_k, \text{sign } \alpha_k)$. The set of voters is

infinite, indexed by (ω, b, s) , and the distribution from which the frequencies are drawn (the Fourier transform of the kernel, times the uniform distribution of b on $[0, 2\pi]$) plays the role of a *prior*: this is the viewpoint of [Letarte et al. \(2019\)](#) discussed in [Section 7.4](#).

Remark. The exercise shows that the question “which predictor?” reduces, for all these methods, to the question “which distribution Q over voters?”, which is precisely the object of PAC-Bayes.

Exercise 1.2

Show that the empirical Rademacher complexity of $\text{conv}(\mathcal{H})$ equals the one of $\mathcal{H}$. Hint: for a fixed vector of signs σ , the function $f \mapsto \frac{1}{m} \sum_i \sigma_i f(x_i)$ is linear in f .

Solution. Recall that $\hat{\mathfrak{R}}_S(\mathcal{F}) = \mathbb{E}_\sigma [\sup_{f \in \mathcal{F}} \frac{1}{m} \sum_i \sigma_i f(x_i)]$, where the σ_i are independent uniform signs. Since $\mathcal{H} \subseteq \text{conv}(\mathcal{H})$, the supremum over the convex hull is at least the supremum over $\mathcal{H}$, so $\hat{\mathfrak{R}}_S(\mathcal{H}) \leq \hat{\mathfrak{R}}_S(\text{conv}(\mathcal{H}))$.

For the converse, fix σ and let $L_\sigma(f) = \frac{1}{m} \sum_i \sigma_i f(x_i)$, which is linear in f . Any $f \in \text{conv}(\mathcal{H})$ can be written $f = \sum_{j=1}^k Q_j h_j$ with $h_j \in \mathcal{H}$, $Q_j \geq 0$ and $\sum_j Q_j = 1$. By linearity,

$$L_\sigma(f) = \sum_j Q_j L_\sigma(h_j) \leq \max_j L_\sigma(h_j) \leq \sup_{h \in \mathcal{H}} L_\sigma(h).$$

Taking the supremum over f and then the expectation over σ gives $\hat{\mathfrak{R}}_S(\text{conv}(\mathcal{H})) \leq \hat{\mathfrak{R}}_S(\mathcal{H})$, hence equality. (For infinite convex combinations the same argument applies with integrals.)

Remark. Averaging voters does not increase the capacity measured by the Rademacher complexity. This is the key to the margin bounds of [Schapire et al. \(1998\)](#) and [Koltchinskii and Panchenko \(2002\)](#), which do not depend on the number of voters in the vote.

Exercise 1.3

In [Fig. 2](#), the training error reaches zero after a few dozen rounds. Using the right panel, explain why a bound based on the empirical margin distribution can keep decreasing afterwards.

Solution. A margin bound ([Section 5.9](#)) has the form $R_{\mathcal{D}}(B_Q) \leq \mathbb{P}_S(M_Q \leq \theta) + O(\sqrt{\ln |\mathcal{H}| \ln m / (m\theta^2)})$ for every $\theta > 0$. Its empirical term is not the training error, which is $\mathbb{P}_S(M_Q \leq 0)$, but the proportion of training examples whose normalized margin is below θ . Once the training error is zero, AdaBoost keeps increasing the margins of the examples: the right panel of [Fig. 2](#) shows that the empirical distribution function of the margins moves to the right between $T = 50$ and $T = 400$ rounds. For a fixed θ , the quantity $\mathbb{P}_S(M_Q \leq \theta)$ therefore keeps decreasing; equivalently, we can use a larger θ for the same empirical term, which decreases the complexity term in $1/\theta^2$. The bound thus keeps improving although the training error no longer moves, which is consistent with the stable (and even slightly decreasing) test error of the middle panel.

Section 2 – Mathematical toolbox

Exercise 2.1

Compute $\text{kl}(0.1\|0.2)$ and $\text{kl}(0.2\|0.1)$. Is the binary kl symmetric?

Solution. By definition $\text{kl}(q\|p) = q \ln \frac{q}{p} + (1-q) \ln \frac{1-q}{1-p}$. Hence

$$\begin{aligned}\text{kl}(0.1\|0.2) &= 0.1 \ln \frac{1}{2} + 0.9 \ln \frac{9}{8} = -0.0693 + 0.1060 = 0.0367, \\ \text{kl}(0.2\|0.1) &= 0.2 \ln 2 + 0.8 \ln \frac{8}{9} = 0.1386 - 0.0942 = 0.0444.\end{aligned}$$

The two values differ: the binary kl is not symmetric. The interpretation of [Section 2](#) helps: $\text{kl}(q\|p)$ measures the surprise of observing a frequency q when the true rate is p . Observing 0.1 errors when the true rate is 0.2 is less surprising than observing 0.2 when the true rate is 0.1, because a Bernoulli variable of mean 0.2 fluctuates more than one of mean 0.1.

Exercise 2.2

Show that the Gibbs distribution Q_ϕ of Lemma [2.8](#) is the only posterior for which the change of measure is an equality.

Solution. The proof of Lemma [2.8](#) shows that, for every Q with $\text{KL}(Q\|P) < \infty$,

$$\text{KL}(Q\|P) + \ln \mathbb{E}_P e^\phi - \mathbb{E}_Q \phi = \text{KL}(Q\|Q_\phi).$$

The change of measure is therefore an equality iff $\text{KL}(Q\|Q_\phi) = 0$. By Gibbs' inequality (the KL is non-negative and vanishes only for equal distributions, see TD Exercise 1), this happens iff $Q = Q_\phi$. If $\text{KL}(Q\|P) = +\infty$, the right-hand side of [\(2.2\)](#) is infinite while the left-hand side is finite (when $\mathbb{E}_Q \phi$ is), so there is no equality either. Hence Q_ϕ is the unique posterior achieving equality, and the supremum in [\(2.3\)](#) is attained only at Q_ϕ .

Exercise 2.3

Using the Occam bound in Pinsker form, how many examples are needed to certify a risk below 0.15 for a classifier with empirical risk 0.1 chosen among 10^6 candidates, with $\delta = 0.05$?

Solution. With the uniform prior on $|\mathcal{H}| = 10^6$ classifiers, $\ln \frac{1}{P(h)} = \ln 10^6 = 13.82$, and $\ln \frac{1}{\delta} = \ln 20 = 3.00$. The Pinsker form of Theorem [2.9](#) gives $R_{\mathcal{D}}(h) \leq 0.1 + \sqrt{\frac{13.82+3.00}{2m}}$. The certificate is below 0.15 iff

$$\sqrt{\frac{16.81}{2m}} \leq 0.05 \iff m \geq \frac{16.81}{2 \times 0.0025} = 3362.2,$$

so $m = 3363$ examples suffice. *Remarks.* With the kl form, far fewer examples are needed: about 1550, since the kl is much tighter for small risks. And the dependence on the number of candidates is only logarithmic: with 10^{12} candidates, the numerator becomes $27.63 + 3.00$ and we need 6127 examples, less than twice as many.

Exercise 2.4

Using Pinsker's inequality, show that the kl interval of Fig. 7 is never wider than the Hoeffding interval. Using notebook 1, determine for which empirical risks it is at least twice narrower when $m = 1000$.

Solution. Let $\varepsilon = \ln(1/\delta)/m$. The Hoeffding interval is $[\hat{q}, \hat{q} + \sqrt{\varepsilon/2}]$ and the kl interval is $[\hat{q}, \bar{\text{kl}}^{-1}(\hat{q}, \varepsilon)]$. If p belongs to the kl interval, then $\text{kl}(\hat{q}||p) \leq \varepsilon$, and Pinsker's inequality gives $2(p - \hat{q})^2 \leq \text{kl}(\hat{q}||p) \leq \varepsilon$, i.e. $p \leq \hat{q} + \sqrt{\varepsilon/2}$. Hence $\bar{\text{kl}}^{-1}(\hat{q}, \varepsilon) \leq \hat{q} + \sqrt{\varepsilon/2}$: the kl interval is always contained in the Hoeffding interval.

Numerically, for $m = 1000$ and $\delta = 0.05$, the Hoeffding width is $\sqrt{\ln 20/2000} = 0.0387$. Computing $\bar{\text{kl}}^{-1}(\hat{q}, \varepsilon) - \hat{q}$ on a grid of $\hat{q}$, the ratio of the two widths is 3.97 at $\hat{q} = 0.01$, still 2 at $\hat{q} \approx 0.054$, and decreases to 1.00 at $\hat{q} = 0.5$. The kl interval is thus at least twice narrower for all empirical risks below about 5.4%. Pinsker's inequality is tight around $\frac{1}{2}$, which is why the two intervals coincide there.

Section 3 – PAC-Bayesian generalization bounds

Exercise 3.1

Starting from Theorem 3.1 with $\Delta(q, p) = 2(q - p)^2$, prove McAllester's bound directly, without going through the kl.

Solution. The function $\Delta(q, p) = 2(q - p)^2$ is jointly convex (it is a convex function of the linear form $q - p$), so Theorem 3.1 applies. It remains to bound $\mathcal{I}_\Delta(m) = \mathbb{E}_{S'} \mathbb{E}_{h \sim P} e^{2m(\hat{R}_{S'}(h) - R_D(h))^2}$. For a fixed h , by Pinsker's inequality, $2(\hat{p} - p)^2 \leq \text{kl}(\hat{p}||p)$, so $\mathbb{E}_{S'} e^{2m(\hat{p} - p)^2} \leq \mathbb{E}_{S'} e^{m \text{kl}(\hat{p}||p)} \leq 2\sqrt{m}$ by Maurer's lemma (Lemma 2.4, for $m \geq 8$). Hence $\mathcal{I}_\Delta(m) \leq 2\sqrt{m}$ and, with probability at least $1 - \delta$, for all Q ,

$$2(\hat{R}_S(G_Q) - R_D(G_Q))^2 \leq \frac{\text{KL}(Q||P) + \ln \frac{2\sqrt{m}}{\delta}}{m},$$

hence

$$R_D(G_Q) \leq \hat{R}_S(G_Q) + \sqrt{\frac{\text{KL}(Q||P) + \ln \frac{2\sqrt{m}}{\delta}}{2m}}.$$

Remark. One can also bound $\mathbb{E} e^{2m(\hat{p} - p)^2}$ without the kl, using the sub-Gaussian tail $\mathbb{P}(|\hat{p} - p| \geq t) \leq 2e^{-2mt^2}$ and the formula $\mathbb{E} e^Z = 1 + \int_0^\infty e^z \mathbb{P}(Z > z) dz$; this gives a bound of order m instead of $2\sqrt{m}$, i.e. a slightly worse constant ($\ln(2m)$ instead of $\ln(2\sqrt{m})$). This is the form of the original bounds of McAllester.

Exercise 3.2

For $m = 500$, $\delta = 0.05$, $\hat{R}_S(G_Q) = 0.05$ and $\text{KL}(Q||P) \in \{0, 5, 20\}$, compute McAllester's and Seeger's bounds, and compare with Table 1.

Solution. The confidence term is $\ln \frac{2\sqrt{500}}{0.05} = \ln 894.4 = 6.80$. The complexity per example $\varepsilon = \frac{\text{KL}+6.80}{500}$ and the two bounds are:

$\text{KL}(Q\ P)$	ε	McAllester $0.05 + \sqrt{\varepsilon/2}$	Seeger $\bar{\text{kl}}^{-1}(0.05, \varepsilon)$
0	0.0136	0.132	0.094
5	0.0236	0.159	0.112
20	0.0536	0.214	0.154

Compared with [Table 1](#) ($m = 1000$, $\hat{R}_S = 0.12$), the gap between the two bounds is larger here: the complexity term of Seeger's bound is about 30% smaller than McAllester's when $\hat{R}_S = 0.12$, but about 40% smaller when $\hat{R}_S = 0.05$. This is the fast rate of the kl at work: the smaller the empirical risk, the more the square root of Pinsker's inequality is wasteful. Halving m roughly doubles ε , which costs Seeger's bound less than McAllester's for the same reason.

Exercise 3.3

Using the linear form of Catoni's bound given after [Corollary 3.4](#), find the value of C that minimizes it when $\hat{R}_S(G_Q) = 0.12$, $\text{KL}(Q\|P) = 5$, $m = 1000$ and $\delta = 0.05$, and compare the result with [Table 1](#).

Solution. Let $A = \frac{\text{KL} + \ln(1/\delta)}{m} = \frac{5+3.00}{1000} = 0.0080$ and $f(C) = \frac{C\hat{R}_S + A}{1 - e^{-C}}$. Setting $f'(C) = 0$ gives $\hat{R}_S(1 - e^{-C}) = (C\hat{R}_S + A)e^{-C}$, i.e. $\hat{R}_S(e^C - 1 - C) = A$. With $\hat{R}_S = 0.12$, $e^C - 1 - C = 0.0667$; since $e^C - 1 - C \approx \frac{C^2}{2} + \frac{C^3}{6}$, a first approximation is $C \approx \sqrt{2A/\hat{R}_S} = 0.365$, and solving numerically gives $C^* = 0.344$ and $f(C^*) = 0.169$. The exact (non-linear) Catoni bound is minimized at $C = 0.371$, where it equals 0.165. In [Table 1](#), Catoni's bound optimized on a grid of 30 values of C (with a union bound, hence $\ln 30$ added to the numerator) gives 0.166 for $\text{KL} = 5$ and Seeger's bound 0.168: the linear relaxation loses only 0.004 with respect to the exact bound, and the grid costs very little. *Warning:* as for λ in the Hoeffding bound, C^* depends on $\hat{R}_S$ and KL , hence on the data; using it directly requires a grid and a union bound.

Exercise 3.4

Reproduce the experiment of [Fig. 10](#) for $m \in \{100, 1000, 10000\}$. Show that the failure frequency of the naive bound does not go to zero when m grows, and explain why by comparing $\sqrt{\ln(1/\delta)/(2m)}$ with the typical deviation of the minimum of 1000 empirical risks.

Solution. We draw 1000 classifiers of true risk 0.5 (their empirical risks are independent $\text{Bin}(m, \frac{1}{2})/m$), keep the smallest empirical risk $\hat{r}_{\min}$, and check whether the naive bound $\hat{r}_{\min} + \sqrt{\ln(1/\delta)/(2m)}$ is below 0.5. Over 2000 repetitions:

m	mean of $0.5 - \hat{r}_{\min}$	naive width	ratio	naive bound wrong	Occam wrong
100	0.160	0.122	1.31	99.8%	0.3%
1000	0.051	0.039	1.32	99.9%	0.5%
10000	0.016	0.012	1.32	100%	0.6%

The failure frequency does not decrease. The reason is that both quantities scale as $1/\sqrt{m}$. Each empirical risk is approximately Gaussian with standard deviation $\frac{1}{2\sqrt{m}}$, and the minimum of n independent Gaussian variables lies about $\sqrt{2\ln n}$ standard deviations below their mean. The selection bias is therefore of order $\frac{\sqrt{2\ln n}}{2\sqrt{m}} = \frac{1.86}{\sqrt{m}}$ for $n = 1000$ (in practice $\approx 1.6/\sqrt{m}$ at these sample sizes), while the naive width is $\frac{\sqrt{\ln(1/\delta)/2}}{\sqrt{m}} = \frac{1.22}{\sqrt{m}}$. Their ratio does not depend on m : the naive bound is always too narrow by the same factor. The Occam bound has width $\frac{\sqrt{(\ln n + \ln(1/\delta))/2}}{\sqrt{m}} = \frac{2.22}{\sqrt{m}}$, which exceeds the selection bias: the price $\ln n$ is exactly what is needed to absorb the selection. More data do not cure selection bias; only accounting for the selection does.

Section 4 – Posteriors, priors and the computation of the bounds

Exercise 4.1

For a finite class and a uniform prior, show that $\text{KL}(Q_\lambda \| P) \rightarrow \ln \frac{|\mathcal{H}|}{k}$ when $\lambda \rightarrow \infty$, where k is the number of empirical risk minimizers.

Solution. Let $n = |\mathcal{H}|$, $r^* = \min_h \hat{R}_S(h)$ and $\mathcal{M} = \{h : \hat{R}_S(h) = r^*\}$, with $|\mathcal{M}| = k$. With the uniform prior,

$$Q_\lambda(h) = \frac{e^{-\lambda m \hat{R}_S(h)}}{\sum_{h'} e^{-\lambda m \hat{R}_S(h')}} = \frac{e^{-\lambda m (\hat{R}_S(h) - r^*)}}{k + \sum_{h' \notin \mathcal{M}} e^{-\lambda m (\hat{R}_S(h') - r^*)}}.$$

For $h' \notin \mathcal{M}$, $\hat{R}_S(h') - r^* > 0$, so the exponentials tend to 0 when $\lambda \rightarrow \infty$. Hence $Q_\lambda(h) \rightarrow \frac{1}{k}$ for $h \in \mathcal{M}$ and $Q_\lambda(h) \rightarrow 0$ otherwise: Q_λ converges to the uniform distribution $U_{\mathcal{M}}$ on the minimizers. The KL is a continuous function of Q on the simplex (with $0 \ln 0 = 0$) when P has full support, so $\text{KL}(Q_\lambda \| P) \rightarrow \text{KL}(U_{\mathcal{M}} \| P) = \sum_{h \in \mathcal{M}} \frac{1}{k} \ln \frac{1/k}{1/n} = \ln \frac{n}{k}$ (TD Exercise 1). With a unique minimizer we recover the Occam price $\ln n$; when several classifiers tie, the Gibbs posterior spreads over them and pays less.

Exercise 4.2

Explain why using the same sample to learn the centre of a Gaussian prior and to compute the empirical term of the bound is not allowed. What happens if we do it anyway?

Solution. The only probabilistic step of the proof of Theorem 3.1 is the third one, where we write $\mathbb{E}_{S'} \mathbb{E}_{h \sim P} e^{m \Delta(\hat{R}_{S'}(h), R_{\mathcal{D}}(h))} = \mathbb{E}_{h \sim P} \mathbb{E}_{S'} e^{m \Delta(\dots)}$ and bound the inner expectation for each fixed h . This requires P not to depend on S' : if $P = P_S$ is centred on a classifier learned on S , the voters drawn from P_S are correlated with the sample, their empirical risks are biased downwards, and the inner expectation can no longer be controlled by Maurer's lemma. The theorem simply does not apply.

If we do it anyway, the “certificate” can be badly wrong. Take the extreme case where the posterior equals the prior, both centred on the learned classifier: the KL is

zero, and the bound becomes a concentration inequality for a classifier that was selected with the data—exactly the naive bound of Fig. 10, which fails almost surely when the selection is strong. With a flexible model (for instance a neural network that interpolates the data), the empirical risk is close to zero, the KL is zero, and the certificate would claim a near-zero risk regardless of the true one. The correct procedure is to split S : learn the prior on S_1 and compute the bound on S_2 (Proposition 6.5, Fig. 15).

Exercise 4.3

In the Monte Carlo certificate, how should δ and δ' be split to minimize the final bound when $N = 100$? when $N = 10^4$?

Solution. The final certificate holds with probability $1 - \delta_{\text{tot}}$ with $\delta_{\text{tot}} = \delta + \delta'$; we fix $\delta_{\text{tot}} = 0.05$ and choose the fraction $\alpha = \delta'/\delta_{\text{tot}}$ given to the Monte Carlo step. Increasing α makes the Monte Carlo inversion $\bar{\text{kl}}^{-1}(\hat{r}_N, \ln(2/\delta')/N)$ tighter but the PAC-Bayesian bound looser (it uses $\delta = (1 - \alpha)\delta_{\text{tot}}$). The two terms enter only through logarithms, so the optimum is flat. On the example of Fig. 16 (breast cancer, $\text{KL} = 32$, $m = 398$), a grid search over α gives the following. With $N = 100$ samples, the Monte Carlo term is the main source of looseness, since $\ln(2/\delta')/N$ is large; it deserves most of the budget, and the optimum is $\alpha \approx 0.86$ ($\delta' = 0.043$, $\delta = 0.007$), with a certificate of 0.380 against 0.386 for the even split. With $N = 10^4$ samples, the Monte Carlo term is small and the PAC-Bayesian term dominates; the optimum moves to $\alpha \approx 0.3$, and the certificate (0.2591) is practically the same as with the even split (0.2594). In practice the even split is a sensible default; the gain from optimizing is small, and the split must anyway be chosen before computing the bound (or on a grid with a union bound). It is more effective to increase N .

Section 5 – From the Gibbs classifier to the majority vote

Exercise 5.1

Give an example of three voters and a distribution on three points for which $R_{\mathcal{D}}(B_Q) = 0$ while $R_{\mathcal{D}}(G_Q) = \frac{1}{3}$. Compute the three oracle bounds.

Solution. Let $\mathcal{D}$ be uniform on three points z_1, z_2, z_3 , and let voter h_i err on z_i only; Q is uniform on $\{h_1, h_2, h_3\}$. On each point exactly one voter out of three errs, so $W_Q(z_j) = \frac{1}{3}$ for all j . The vote is correct everywhere ($W_Q < \frac{1}{2}$): $R_{\mathcal{D}}(B_Q) = 0$, while $R_{\mathcal{D}}(G_Q) = \mathbb{E}W_Q = \frac{1}{3}$. The second-order quantities are $e_Q = \mathbb{E}W_Q^2 = \frac{1}{9}$ and $d_Q = \mathbb{E}[2W_Q(1 - W_Q)] = 2 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{9}$. Hence:

$$2R_{\mathcal{D}}(G_Q) = \frac{2}{3}, \quad 4e_Q = \frac{4}{9}, \quad C_Q = 1 - \frac{(1 - 2/3)^2}{1 - 8/9} = 1 - \frac{1/9}{1/9} = 0$$

for the first-order bound, the tandem bound and the C-bound respectively. The C-bound is exact here: W_Q is constant, its variance is zero, and Cantelli's inequality says that a variable with zero variance equal to $\frac{1}{3}$ is never above $\frac{1}{2}$. The example is a small Condorcet jury with perfectly “anti-correlated” errors: the disagreement ($\frac{4}{9}$) exceeds

the Gibbs risk ($\frac{1}{3}$), so Proposition 5.7 predicts the order C-bound $\leq$ tandem $\leq$ first order, as observed.

Exercise 5.2

Show that the C-bound is always smaller than 1 when $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$, and that it equals $4R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q))$ when $d_{\mathcal{D}}(Q) = 0$. Compare with the first-order bound.

Solution. Write $R = R_{\mathcal{D}}(G_Q)$ and $d = d_{\mathcal{D}}(Q)$. By Proposition 5.3, $0 \leq d \leq 2R(1 - R) < \frac{1}{2}$, so $1 - 2d > 0$. When $R < \frac{1}{2}$, $(1 - 2R)^2 > 0$, hence $\frac{(1-2R)^2}{1-2d} > 0$ and $\mathcal{C}_Q < 1$. (In fact $1 - 2d \geq 1 - 4R + 4R^2 = (1 - 2R)^2$, so the ratio is at most 1 and $\mathcal{C}_Q \geq 0$.) When $d = 0$, $\mathcal{C}_Q = 1 - (1 - 2R)^2 = 4R - 4R^2 = 4R(1 - R)$. Compared with the first-order bound $2R$: $4R(1 - R) \geq 2R \iff 2(1 - R) \geq 1 \iff R \leq \frac{1}{2}$. So when the voters never disagree, the C-bound is *worse* than the first-order bound, by a factor $2(1 - R)$ which is close to 2 for small R . This is consistent with Proposition 5.7 ($d = 0 < R$): without diversity, second-order information can only hurt, since all voters predict the same thing and the vote is just one of them.

Exercise 5.3

(*) Prove that the minimum over μ of the Chebyshev–Cantelli family equals the C-bound (TD, Exercise 5).

Solution. Let $R = \mathbb{E}W_Q < \frac{1}{2}$, $e = \mathbb{E}W_Q^2$, $v = e - R^2 = \text{Var } W_Q$, and $f(\mu) = \frac{e - 2\mu R + \mu^2}{(1/2 - \mu)^2}$ for $\mu < \frac{1}{2}$. The numerator equals $\mathbb{E}(W_Q - \mu)^2 = v + (R - \mu)^2$. Differentiating,

$$f'(\mu) = \frac{2(\mu - R)(\frac{1}{2} - \mu) + 2(v + (R - \mu)^2)}{(1/2 - \mu)^3} = \frac{2[v - (R - \mu)(\frac{1}{2} - R)]}{(1/2 - \mu)^3},$$

using $(\mu - R)(\frac{1}{2} - \mu) + (R - \mu)^2 = (R - \mu)(R - \frac{1}{2})$. The bracket is increasing in μ and vanishes at $\mu^* = R - \frac{v}{1/2 - R} = \frac{R/2 - e}{1/2 - R}$; f decreases before μ^* and increases after, so μ^* is the minimum (and $\mu^* < R < \frac{1}{2}$ is admissible). At μ^* , $R - \mu^* = \frac{v}{a}$ with $a = \frac{1}{2} - R$, and $\frac{1}{2} - \mu^* = a + \frac{v}{a} = \frac{a^2 + v}{a}$. Hence

$$f(\mu^*) = \frac{v + v^2/a^2}{(a^2 + v)^2/a^2} = \frac{v(a^2 + v)}{(a^2 + v)^2} = \frac{v}{v + a^2}.$$

Finally, with the margin $M_Q = 1 - 2W_Q$: $v = \frac{1}{4} \text{Var } M_Q$ and $a^2 = \frac{1}{4}(\mathbb{E}M_Q)^2$, so $f(\mu^*) = \frac{\text{Var } M_Q}{\text{Var } M_Q + (\mathbb{E}M_Q)^2} = 1 - \frac{(\mathbb{E}M_Q)^2}{\mathbb{E}M_Q^2} = \mathcal{C}_Q$, the C-bound (5.4). The tandem bound is $f(0)$, hence always at least the C-bound, which proves the first claim of Proposition 5.7.

Exercise 5.4

In Condorcet's model with n independent voters of error p , compute $e_{\mathcal{D}}(Q)$ and $d_{\mathcal{D}}(Q)$, and show that the C-bound decreases like $1/n$ while the tandem bound tends to $4p^2$.

Solution. Since $nW_Q \sim \text{Bin}(n, p)$, $\mathbb{E}W_Q = p$ and $\text{Var } W_Q = \frac{p(1-p)}{n}$. Hence $e_{\mathcal{D}}(Q) = \mathbb{E}W_Q^2 = p^2 + \frac{p(1-p)}{n}$ and, by Proposition 5.3, $d_{\mathcal{D}}(Q) = 2(p - e_{\mathcal{D}}(Q)) = 2p(1-p)(1 - \frac{1}{n})$. The tandem bound is $4e_{\mathcal{D}}(Q) = 4p^2 + \frac{4p(1-p)}{n} \rightarrow 4p^2$: it never goes to zero, even though the risk of the vote does (for $p = 0.3$ it tends to 0.36). For the C-bound,

$$1 - 2d_{\mathcal{D}}(Q) = 1 - 4p(1-p) + \frac{4p(1-p)}{n} = (1-2p)^2 + \frac{4p(1-p)}{n},$$

$$C_Q = 1 - \frac{(1-2p)^2}{1 - 2d_{\mathcal{D}}(Q)} = \frac{4p(1-p)/n}{(1-2p)^2 + 4p(1-p)/n}.$$

When $n \rightarrow \infty$, $C_Q \sim \frac{4p(1-p)}{(1-2p)^2} \cdot \frac{1}{n}$. For $p = 0.3$ the equivalent is $\frac{0.84}{0.16n} = \frac{5.25}{n}$; the exact values are 0.174 for $n = 25$ and 0.049 for $n = 101$ (Fig. 18). The true risk decreases exponentially fast (by Hoeffding's inequality, $R_{\mathcal{D}}(B_Q) \leq e^{-2n(1/2-p)^2}$), so even the C-bound is pessimistic, but it captures the qualitative effect of independence. The tandem bound misses it because it applies Markov's inequality to W_Q^2 , i.e. it measures the deviations of W_Q from 0: since W_Q concentrates around $p > 0$, $\mathbb{E}W_Q^2 \rightarrow p^2$ does not vanish. The C-bound measures the deviations from the well-chosen centre μ^* (Exercise 5.3), and only pays for the variance $\frac{p(1-p)}{n}$, which vanishes.

Section 6 – Self-bounding learning

Exercise 6.1

Reproduce the experiment of Fig. 10 (draw 1000 random classifiers and $m = 100$ labels). How often is the “fixed- h ” Hoeffding bound violated? And the Occam bound? What happens to the Occam bound with 10^6 classifiers?

Solution. See Exercise 3.4 for the code and the explanation. With 1000 classifiers and $m = 100$, the best empirical risk is 0.34 on average; the fixed- h Hoeffding bound ($+0.122$) is violated in about 99.8% of the runs, while the Occam bound ($+\sqrt{(\ln 1000 + \ln 20)/200} = +0.222$) is violated in about 0.3% of them, below the nominal 5%. With 10^6 classifiers, the best empirical risk drops to 0.26 on average (a stronger selection), but the Occam price grows only to $\sqrt{(\ln 10^6 + \ln 20)/200} = 0.290$: the certificate, about 0.55 on average, is never violated in 200 runs. It is vacuous, which is the honest answer: with 100 examples one cannot distinguish a million random classifiers from each other. The logarithmic price keeps the bound valid for any number of candidates, at the cost of informativeness.

Exercise 6.2

Show that the value of λ minimizing the Hoeffding-type bound (3.5) depends on $\text{KL}(Q\|P)$, and explain why using it directly is forbidden, whereas using λ^ in the PAC-Bayes- λ bound is allowed.*

Solution. The bound is $\widehat{R}_S(G_Q) + \frac{A}{\lambda} + \frac{\lambda}{8}$ with $A = \frac{\text{KL}(Q\|P) + \ln(1/\delta)}{m}$; its derivative in λ vanishes at $\lambda^* = \sqrt{8A} = \sqrt{8(\text{KL}(Q\|P) + \ln \frac{1}{\delta})/m}$, which depends on $\text{KL}(Q\|P)$, hence on the posterior and on the data. In Corollary 3.5, λ is a parameter of the comparison function $\Delta(q, p) = \lambda(p - q)$: each value of λ defines a different event of probability at least $1 - \delta$, and the corollary guarantees only that the event associated with a λ fixed in advance holds. Choosing λ after seeing the data means picking one event among infinitely many, and the probability that the picked one fails is not controlled (the situation of Fig. 10). It must be done on a finite grid with a union bound (Proposition 6.3).

In Corollary 3.6, on the contrary, the probabilistic event is the single event of Seeger's bound, which does not involve λ ; the PAC-Bayes- λ inequality is derived from it by deterministic algebra valid for every $\lambda \in (0, 2)$ simultaneously. Plugging the data-dependent λ^* is therefore legitimate: the rule is “free if it indexes a family of inequalities all implied by the same event, paid if it changes the event” (Section 6.3).

Exercise 6.3

Using Proposition 6.6, compute $\partial p / \partial q$ at $\hat{q} = 0.1$, $\varepsilon = 0.05$ (with $p \approx 0.220$). Which ingredient of the certificate is it more profitable to decrease: the empirical risk by 0.01, or the KL by 5 nats with $m = 1000$?

Solution. With $p = 0.2201$, $q = 0.1$: $\frac{p(1-p)}{p-q} = \frac{0.2201 \times 0.7799}{0.1201} = 1.429$ and $\ln \frac{p(1-q)}{q(1-p)} = \ln \frac{0.1981}{0.0780} = \ln 2.540 = 0.932$. Hence

$$\frac{\partial p}{\partial q} = 1.429 \times 0.932 = 1.33, \quad \frac{\partial p}{\partial \varepsilon} = 1.43.$$

Decreasing the empirical risk by 0.01 decreases the certificate by about $1.33 \times 0.01 = 0.0133$ (the exact value is 0.0135). Decreasing the KL by 5 nats decreases ε by $5/1000 = 0.005$ and the certificate by about $1.43 \times 0.005 = 0.0071$ (exact: 0.0073). At this operating point, one point of empirical risk is worth about 9.3 nats of KL: it is more profitable to improve the fit. The balance changes with m : for $m = 100$, the same 5 nats would be worth 0.071, five times more than the gain in empirical risk. This is exactly the trade-off that the Gibbs posterior or a gradient descent on the certificate resolves automatically.

Exercise 6.4

(*) *Derive the “quadratic” relaxation of the table of Section 6.6 from the refined Pinsker inequality. Writing $a = \sqrt{\varepsilon/2}$, show that it is smaller than McAllester's bound iff $2a + 2\sqrt{\hat{q} + a^2} \leq 1$; in particular, for small ε it is the case as soon as $\hat{q} < \frac{1}{4}$.*

Solution. Let $p = \bar{\text{kl}}^{-1}(\hat{q}, \varepsilon) > \hat{q}$. By the refined Pinsker inequality, $\varepsilon \geq \text{kl}(\hat{q} \| p) \geq \frac{(p - \hat{q})^2}{2p}$, so $(p - \hat{q})^2 \leq 2\varepsilon p$, i.e. $p^2 - 2(\hat{q} + \varepsilon)p + \hat{q}^2 \leq 0$. The roots of this quadratic are $\hat{q} + \varepsilon \pm \sqrt{2\hat{q}\varepsilon + \varepsilon^2}$, so

$$p \leq \hat{q} + \varepsilon + \sqrt{2\hat{q}\varepsilon + \varepsilon^2} = \left(\sqrt{\hat{q} + \frac{\varepsilon}{2}} + \sqrt{\frac{\varepsilon}{2}} \right)^2,$$

where the last equality is checked by expanding: $(\sqrt{\hat{q} + a^2} + a)^2 = \hat{q} + 2a^2 + 2a\sqrt{\hat{q} + a^2}$ with $2a^2 = \varepsilon$ and $2a\sqrt{\hat{q} + a^2} = \sqrt{2\varepsilon}\sqrt{\hat{q} + \varepsilon/2} = \sqrt{2\hat{q}\varepsilon + \varepsilon^2}$. McAllester's bound is $\hat{q} + a$. Then

$$(\sqrt{\hat{q} + a^2} + a)^2 \leq \hat{q} + a \iff 2a^2 + 2a\sqrt{\hat{q} + a^2} \leq a \iff 2a + 2\sqrt{\hat{q} + a^2} \leq 1$$

(dividing by $a > 0$). When $\varepsilon \rightarrow 0$, $a \rightarrow 0$ and the condition becomes $2\sqrt{\hat{q}} < 1$, i.e. $\hat{q} < \frac{1}{4}$. So the quadratic bound improves on McAllester's for all classifiers better than 25% error, provided the complexity term is small; for $\hat{q}$ close to $\frac{1}{2}$, Pinsker's inequality is tight and McAllester's bound is better than this relaxation (but never better than the kl itself).

Section 7 – Self-bounding algorithms for ensembles and beyond

Exercise 7.1

Derive the gradient (7.3) of the tandem objective, and check it numerically with finite differences (notebook 4).

Solution. Write $c_1 = \frac{4}{1-\lambda/2}$ and $c_2 = \frac{4}{\lambda(1-\lambda/2)n_0}$, so that $\mathcal{B}_{\text{TND}}(\rho) = c_1 \rho^\top \hat{\mathbf{L}} \rho + c_2 (2 \text{KL}(\rho \| \pi) + \ln \frac{2\sqrt{n_0}}{\delta})$. The matrix $\hat{\mathbf{L}}$ of tandem losses is symmetric ($\hat{L}_{tt'} = \hat{L}_{t't}$), so $\nabla_\rho(\rho^\top \hat{\mathbf{L}} \rho) = 2\hat{\mathbf{L}}\rho$. For the KL, $\text{KL}(\rho \| \pi) = \sum_t \rho_t \ln \rho_t - \rho_t \ln \pi_t$, and $\partial_{\rho_t} = \ln \rho_t + 1 - \ln \pi_t$, i.e. $\nabla_\rho \text{KL} = \ln \frac{\rho}{\pi} + \mathbf{1}$ (componentwise). Hence

$$\nabla_\rho \mathcal{B}_{\text{TND}} = 4 \left[\frac{2\hat{\mathbf{L}}\rho}{1 - \lambda/2} + \frac{2(\ln \frac{\rho}{\pi} + \mathbf{1})}{\lambda(1 - \lambda/2)n_0} \right].$$

With the softmax parametrization $\rho = \text{softmax}(\theta)$, the Jacobian is $\frac{\partial \rho_s}{\partial \theta_t} = \rho_s (\mathbf{1}[s = t] - \rho_t)$, so $\nabla_\theta \mathcal{B} = \rho \odot (g - (\rho^\top g) \mathbf{1})$ with $g = \nabla_\rho \mathcal{B}$, which is (6.2). (The constant $\mathbf{1}$ in the KL gradient disappears in this projection, as it should: it corresponds to the direction that changes $\sum_t \rho_t$.) *Numerical check.* On the forest of the running example ($n_0 = 423$), at random logits θ and $\lambda = 0.7$, the centred finite differences with step 10^{-6} agree with the analytical gradient up to a relative error of 6×10^{-8} :

```
num = np.array([(f(th + 1e-6*e) - f(th - 1e-6*e)) / 2e-6
                 for e in np.eye(100)]])
print(np.linalg.norm(g - num) / np.linalg.norm(num))    # 5.6e-08
```

Exercise 7.2

Explain why the uniform prior over the trees of a random forest would not be a valid prior if the empirical losses were computed on the whole training sample.

Solution. The uniform prior on the *indices* of the trees is fixed in advance, and in this sense it is legitimate. The problem lies in the third step of the proof of Theorem 3.1: for each index t , we need the empirical loss $\hat{L}_t$ to be an average of i.i.d. variables of mean $R_{\mathcal{D}}(h_t)$, independent of h_t . If $\hat{L}_t$ is computed on the whole training sample, it includes the examples used to grow h_t , on which a fully grown tree makes almost no error: $\hat{L}_t$ is then a strongly biased estimate of $R_{\mathcal{D}}(h_t)$ (close to 0 for unpruned trees), and Maurer's lemma does not apply. Equivalently, the prior on the trees h_t themselves (not on their indices) depends on the sample used to evaluate them. The certificate would be meaningless: it would certify a near-zero risk for a forest whose trees err on 19% of the test digits. The out-of-bag construction restores the independence by evaluating each tree only on the examples it has not seen (Theorem 7.1).

Exercise 7.3

In PBGD, show that when all the training points have a large normalized margin, the gradient of the loss term vanishes, and interpret the resulting value of $\|\mathbf{w}\|$.

Solution. The gradient of the objective is $\nabla F(\mathbf{w}) = -C \sum_i \varphi(t_i) \frac{y_i x_i}{\|x_i\|} + \mathbf{w}$ with $t_i = \frac{y_i \langle \mathbf{w}, x_i \rangle}{\|x_i\|}$ and φ the standard Gaussian density. Since $\|\frac{y_i x_i}{\|x_i\|}\| = 1$, the loss part has norm at most $C \sum_i \varphi(t_i) \leq Cm \varphi(t_{\min})$, and $\varphi(t) = \frac{e^{-t^2/2}}{\sqrt{2\pi}} \rightarrow 0$ very fast: if all $t_i \geq t_{\min}$ large, the loss gradient vanishes. Then $\nabla F(\mathbf{w}) \approx \mathbf{w} \neq 0$, and gradient descent shrinks $\mathbf{w}$. Shrinking $\mathbf{w}$ decreases all normalized margins proportionally (t_i is linear in $\|\mathbf{w}\|$), until the smallest ones become of order 1–2, where $\varphi(t_i)$ is no longer negligible. At a stationary point, $\mathbf{w} = C \sum_i \varphi(t_i) \frac{y_i x_i}{\|x_i\|}$: $\mathbf{w}$ is a combination of the points with moderate margins, the analogue of support vectors. If the data are separable with geometric normalized margin γ in direction $\mathbf{u}$, the stationary $\mathbf{w} = s\mathbf{u}$ has $s\gamma = O(1)$, i.e. $\|\mathbf{w}\| \approx \frac{c}{\gamma}$ with c growing slowly (logarithmically) with C , since $\varphi(t)$ decays like $e^{-t^2/2}$. The KL, $\frac{1}{2}\|\mathbf{w}\|^2 \approx \frac{c^2}{2\gamma^2}$, is thus inversely proportional to the squared margin: large margins give small certificates, which is the PAC-Bayesian explanation of the SVM. On the example of Fig. 31, the norm grows from 3.5 ($C = 1$) to 13.6 ($C = 100$) only because of the noisy points, which pull $\mathbf{w}$ as long as their margin is moderate.

Exercise 7.4

With notebook 4, compute the test disagreement of the C-bound posterior of Table 3 and check whether the condition $d \geq R$ of Proposition 5.7 holds. Why is its certificate nevertheless larger than the tandem certificate?

Solution. For the posterior minimizing the PAC-Bayesian C-bound, the test Gibbs risk is $R = 0.176$ and the test disagreement is $d = 0.246$: the condition $d \geq R$ holds, and indeed the *oracle* C-bound is 0.171, smaller than the oracle tandem bound $4e = 0.210$.

Its certificate, however, is 0.531, against 0.361 for the TND-optimized posterior. The reason is statistical. The certificate replaces R by an upper confidence bound and d by a lower confidence bound, each with $\delta/2$: from the OOB estimates $\hat{R} = 0.153$ ($n_1 = 747$) and $\hat{d} = 0.224$ ($n_3 = 423$ pairs), with $\text{KL} = 0.73$, we get $\bar{r} = 0.212$ and $\underline{d} = 0.147$. The disagreement is estimated on the smaller pairwise sample with a doubled KL, and it enters through $\frac{1}{1-2d}$, which amplifies its estimation error: plugging $\bar{r}$ and $\underline{d}$ gives $1 - \frac{(1-0.424)^2}{1-0.294} = 0.531$. The tandem certificate needs a single kl inversion of a small quantity ($\hat{e} = 0.041$), where the kl is at its best. This is the general lesson of Proposition 5.7: the best oracle bound is not necessarily the best certificate; the number of quantities to estimate, and their size, matter.

Exercise 7.5

(*) For a stochastic network with $D = 10^5$ weights, $\sigma_j = \sigma_0 = 0.03$ and $\|\boldsymbol{\mu} - \boldsymbol{\mu}_0\| = 3$, compute $\text{KL}(Q\|P)$. How many examples are needed for the complexity term KL/m to be below 0.01? What does this suggest about data-dependent priors?

Solution. With $\sigma_j = \sigma_0$, the terms $\frac{\sigma_j^2}{\sigma_0^2} - 1 + \ln \frac{\sigma_0^2}{\sigma_j^2}$ vanish, and

$$\text{KL}(Q\|P) = \frac{1}{2} \sum_j \frac{(\mu_j - \mu_{0,j})^2}{\sigma_0^2} = \frac{\|\boldsymbol{\mu} - \boldsymbol{\mu}_0\|^2}{2\sigma_0^2} = \frac{9}{2 \times 0.0009} = 5000 \text{ nats.}$$

The complexity term KL/m is below 0.01 only when $m \geq 5000/0.01 = 500\,000$ examples, about eight times the size of MNIST's training set, and this ignores the confidence term and the kl inversion. Note that the dimension D does not appear: what matters is how far, in units of σ_0 , the posterior mean has travelled from the prior mean. Moving 3 units of Euclidean distance with a noise scale of 0.03 is enormous. This suggests two levers: take σ_0 as large as the network tolerates (chosen on a grid with a union bound), and above all bring $\boldsymbol{\mu}_0$ close to $\boldsymbol{\mu}$ with a data-dependent prior: if $\boldsymbol{\mu}_0$ is a network trained on half of the data, the final network may for instance move by a distance of 0.3 instead of 3, which divides the KL by 100 (50 nats): the certificate then becomes informative with a few tens of thousands of examples. This is the approach of Dziugaite and Roy (2018) and Pérez-Ortiz et al. (2021) (Proposition 6.5).

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